TYLER ROBERTS

Senior Software Engineer

tyler.r.dev@outlook.com | linkedin.com/in/tyler-r-5788a13ba | Florida, United States | +1 (321) 376-7095

SUMMARY

Senior Mobile iOS Engineer with 10+ years of experience delivering consumer, SaaS, and platform-connected applications across the **mobile software industry**, specializing in **Swift 5.0–5.10**, **Objective-C**, **UIKit**, **SwiftUI**, and API-driven product development for scalable iPhone and iPad experiences. Strong track record modernizing legacy Objective-C codebases, improving crash stability, optimizing startup and rendering performance, and shipping secure mobile features tied to backend services, analytics, payments, and real-time workflows. Known for owning architecture, debugging complex production issues, mentoring teams, and turning unstable legacy apps into maintainable, release-ready products.

SKILLS

- **Mobile Development** — **Swift 5.0–5.10**, **Objective-C**, **UIKit**, **SwiftUI**, Auto Layout, Xcode, Interface Builder, Storyboards, programmatic UI, MVVM, MVC, Coordinator pattern, modular architecture, iPhone/iPad app development, App Store deployment
- **Apple Frameworks** — **Combine**, **Core Data**, URLSession, Grand Central Dispatch, OperationQueue, UserDefaults, Keychain, Core Animation, MapKit, AVFoundation, Core Location, Push Notifications, Background Tasks, WidgetKit, deep linking
- **Backend & APIs** — REST APIs, JSON parsing, WebSockets, GraphQL basics, OAuth2, JWT, secure token handling, offline sync, pagination, caching, file upload/download, third-party SDK integration, Firebase, analytics events
- **Data & Persistence** — Core Data, SQLite, Realm, local caching, sync conflict handling, migration strategies, serialization, background persistence, error recovery flows
- **Testing & Quality** — XCTest, XCUITest, unit testing, UI testing, snapshot testing basics, TestFlight, crash analysis, memory leak debugging, Instruments, performance profiling, CI validation, release checklists
- **Cloud & DevOps** — Fastlane, GitHub Actions, Bitrise, Jenkins, Firebase App Distribution, App Store Connect, code signing, provisioning profiles, CI/CD pipelines, feature flag rollout, staged releases
- **Observability & Security** — Crashlytics, Sentry, structured logging, mobile analytics, secure storage, certificate pinning basics, privacy-aware data handling, RBAC-aware client flows, audit-friendly event tracking

EXPERIENCE

Senior Software Engineer

Web Venturez LLC

February 2020 - October 2025

- Led the modernization of a high-traffic iOS product from mixed **Objective-C** modules into a cleaner **Swift 5.7–5.10** architecture, reducing regression risk during releases and improving long-term maintainability for multiple feature teams.
- Built new customer-facing flows in **UIKit** and selected **SwiftUI** surfaces for settings, onboarding, and account experiences, which improved delivery speed by **32%** and reduced UI rework during design revisions.
- Resolved a long-running startup crash tied to race conditions between token refresh, cached profile hydration, and deep-link routing by redesigning launch sequencing and adding guarded async state transitions.
- Improved scrolling performance on feed and dashboard screens by reworking cell rendering, image decoding, and background prefetch behavior, cutting visible frame drops by **41%** on older iPhone models.
- Implemented a safer offline sync model for intermittent-network users by introducing operation queues, retry policies, and clearer reconciliation logic, which lowered duplicate submissions by **67%** in unstable sessions.
- Migrated legacy networking utilities to a unified API client using typed request models, centralized error mapping, and stronger auth handling, making backend contract changes much easier to roll out safely.
- Added secure storage improvements around session persistence using **Keychain** and hardened refresh-token behavior, which reduced forced re-logins by **38%** while strengthening production security posture.
- Diagnosed a memory-growth issue caused by retained closures, notification observers, and image cache misuse, then applied lifecycle cleanup and profiling with Instruments to stabilize long session performance.
- Introduced modular feature boundaries and reusable UI components so that payment, profile, and notification modules could evolve independently without repeatedly breaking shared navigation and state flows.

- Integrated analytics, crash reporting, and release telemetry using **Crashlytics** and structured event tracking, giving product and engineering teams faster visibility into failure patterns after deployments.
- Built CI/CD improvements with **Fastlane** and GitHub-based release automation, reducing manual signing and packaging effort by **55%** and making TestFlight promotion more predictable across environments.
- Supported junior engineers through code reviews, debugging sessions, and architectural guidance, especially around bridging **Objective-C** dependencies into modern **Swift** modules without disruptive rewrites.

Full Stack Engineer

Webiz Digital

July 2017 - January 2020

- Developed and maintained iOS client features in **Swift 4.2–5.5** and legacy **Objective-C**, delivering transactional and account-management workflows that depended on stable REST integrations and reliable local state handling.
- Created reusable **UIKit** components for forms, lists, and detail screens, which accelerated screen delivery by **29%** and improved visual consistency across multiple white-label mobile products.
- Fixed a difficult data-loss issue where background app termination interrupted draft persistence, then redesigned save triggers and recovery behavior so partially completed workflows could be resumed safely.
- Migrated older networking and serialization code from callback-heavy Objective-C helpers into clearer **Swift** service layers, improving readability and reducing onboarding time for new engineers joining the mobile team.
- Enhanced media upload reliability by adding chunk-aware retry handling, progress reporting, and better validation messaging, which reduced failed upload support tickets by **34%** during high-usage periods.
- Implemented push notification flows, deep links, and contextual routing so users could land directly in the correct content state, increasing notification-driven engagement by **21%** after rollout.
- Refactored heavily coupled view controllers into smaller presentation and service layers, solving repeated bugs where one change in a shared screen accidentally broke validation or state restoration elsewhere.
- Worked closely with backend engineers on pagination contracts, error semantics, and response normalization, preventing mobile-specific parsing failures when server payloads changed across environments.
- Introduced crash-monitoring discipline with release tagging, defect triage, and post-release validation checklists, which helped bring crash-free user sessions up by **18%** over successive releases.
- Handled difficult UI issues related to keyboard overlap, Auto Layout ambiguity, and dynamic content height calculations, producing more stable screens across a wider range of device sizes.
- Contributed to release automation and TestFlight distribution workflows, reducing last-minute packaging errors and making handoff from engineering to QA much smoother for recurring mobile releases.

Software Developer

Centric Tech Inc.

October 2013 - June 2017

- Built early iOS business application features using **Objective-C** and **UIKit**, supporting account workflows, data-entry screens, and role-based mobile operations for internal and customer-facing tools.
- Maintained legacy code that relied on Storyboards, delegates, and notification-driven state changes, while gradually introducing cleaner organization patterns that reduced the cost of adding new screens.
- Resolved a persistent issue where rapid navigation and slow API responses produced stale screen states, then added request cancellation and guarded UI refresh logic to prevent misleading customer data displays.
- Improved local persistence and cached state recovery using lightweight storage strategies, which reduced blank-screen and forced-login incidents by **26%** in reconnect and relaunch scenarios.
- Worked on secure login, session timeout, and logout handling, tightening client-side auth behavior and making mobile access patterns align more closely with backend security expectations.
- Refined table rendering, image loading, and view lifecycle handling to improve responsiveness on older devices, cutting screen transition lag by **23%** in commonly used operational flows.
- Supported gradual migration preparation from older Objective-C-only modules toward future **Swift** adoption by separating reusable business logic and reducing direct UI-service coupling.
- Integrated third-party analytics and monitoring SDKs carefully to avoid startup penalties and duplicate events, which improved release diagnostics without introducing measurable user-facing slowdown.

- Fixed multiple production bugs tied to timezone conversion, inconsistent date formatting, and server-side nullability assumptions that previously caused edge-case crashes in reporting screens.
- Collaborated with QA and product teams to reproduce device-specific defects, especially around memory warnings, modal presentation timing, and network recovery, leading to more stable recurring releases.

EDUCATION

Bachelor’s degree in Computer Science

Florida State University

2009 – 2013

- Focused on strong foundations in **software engineering**, **data structures**, and **systems thinking**, which later supported production ownership across mobile apps and cloud-connected services.